Automated Root Tip Inoculation

Using Computer Vision and Robotics

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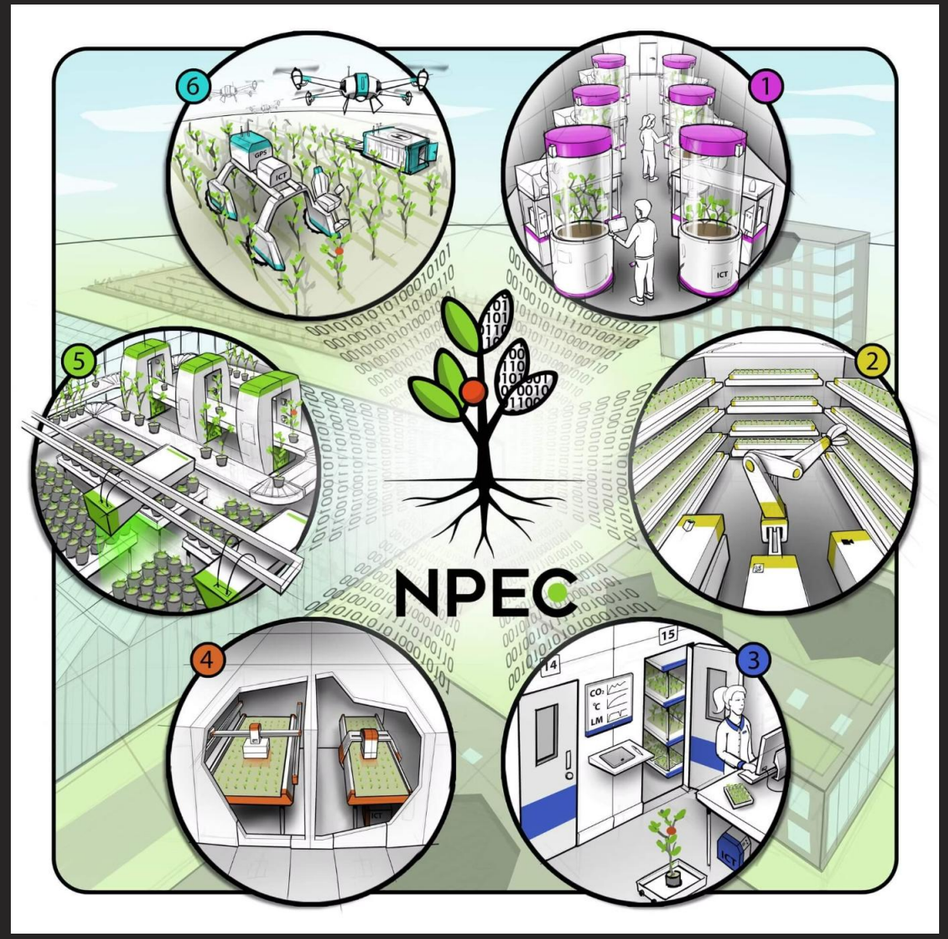
Client: NPEC (Netherlands Plant Eco-phenotyping Centre)

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What is NPEC?

The Challenge

NPEC studies plants. Their system Hades grows over 10,000 seedlings in Petri dishes. Scientists need to place bacteria on each root tip, but doing this by hand is too slow and inconsistent. They need an automated solution.



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Why This Matters

Plant-Microbe Research Find disease-resistant plants

Crop Development Stronger crops, sustainable farming

High-Throughput Automation Hundreds of plates, ~32 seconds each

Impact: Scientists can focus on research analysis rather than spending hours on manual pipetting tasks.

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Complete System Pipeline

1

Image Input

Petri dish photograph

2

CV Pipeline

U-Net segmentation

3

Coordinate Transform

Pixels to robot space

4

PID Controller

Robot positioning

5

Inoculation

Precise delivery

Key Components

- Root segmentation using U-Net deep learning
- Skeleton analysis for tip detection
- Pixel-to-robot coordinate conversion
- PID-controlled robotic pipette

Performance

- 100% success rate
- <1mm accuracy
- 32 seconds per plate
- Fully automated operation

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Dataset Preparation

Data Sources

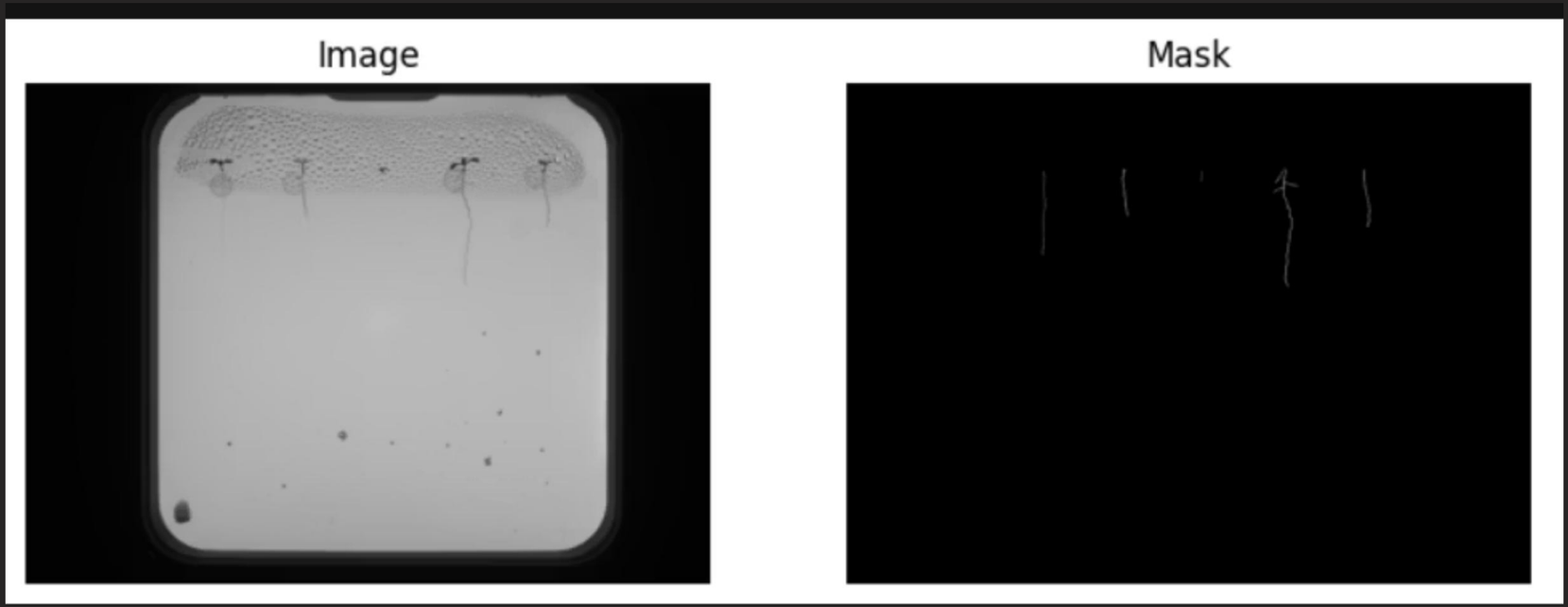
Combined images from three years:

- Y2B_23: 115 images with masks
- Y2B_24: 314 images with masks
- Y2B_25: 17 validation images
- **Total:** 475 training + 17 validation

Pre-processing Pipeline

1. Crop Petri dish using Otsu thresholding
2. Apply CLAHE contrast enhancement
3. Extract 192×192 patches with 50% overlap

Result: 399,475 training patches



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Root Segmentation Model

Architecture & Training

Model: U-Net with 1.9M parameters

Input: 192×192 grayscale patches

Output: Binary root mask

Training: 25 epochs, batch size 16

Learning rate: 1e-4 → 6.25e-6

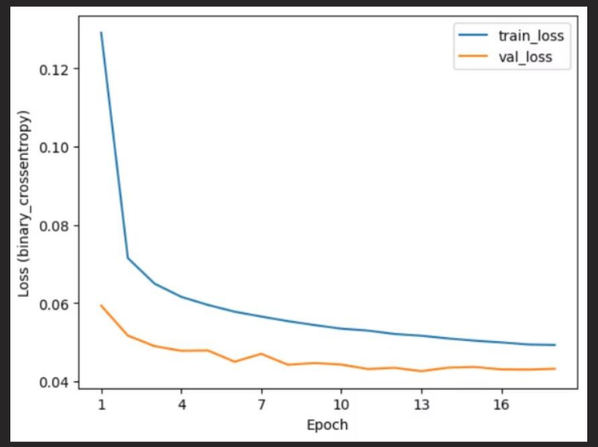
Call backs: Early Stopping

Custom Loss Function

Weighted BCE: pos_weight=3.0 (roots count 3× more)

Focal Tversky: alpha=0.15, beta=0.85, gamma=0.75

- High beta penalises missed roots
- Focal term focuses on hard pixels



~0.85

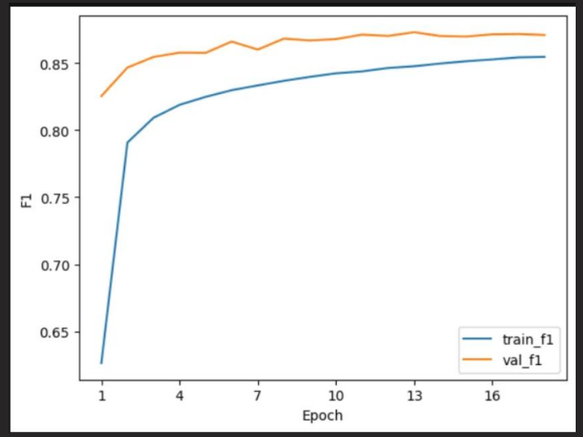
Validation F1 Score

Target: ≥0.8

0.091

Validation Loss

Converged smoothly



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Full Computer Vision Pipeline



20	▼ 1	Ahmed Ali 245226		9.359	23	2mo

Result

sMAPE: 9.36% (<10%)

Rank: 20th place

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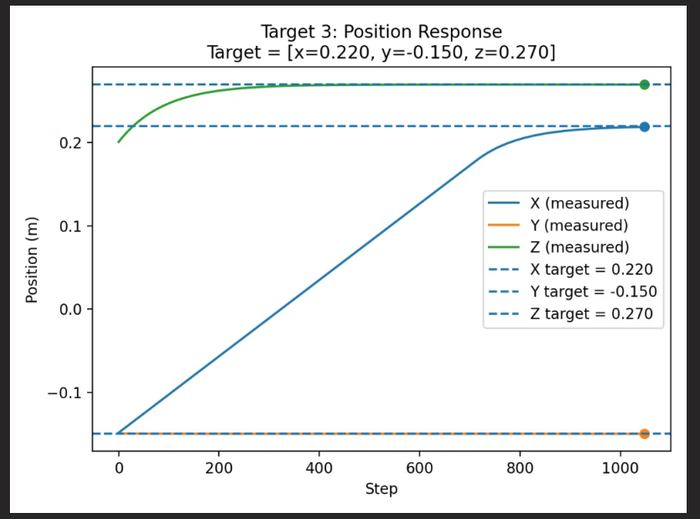
Robotics Controllers

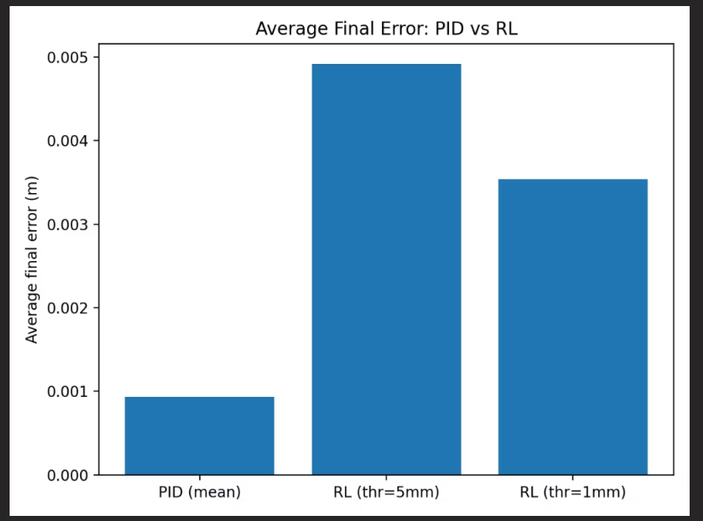
PID Controller:

- 3 controllers (X, Y, Z) with Kp=3.5, Ki=0, Kd=1.2
- Mean error: 0.93mm (<1mm)
- 5/5 targets reached

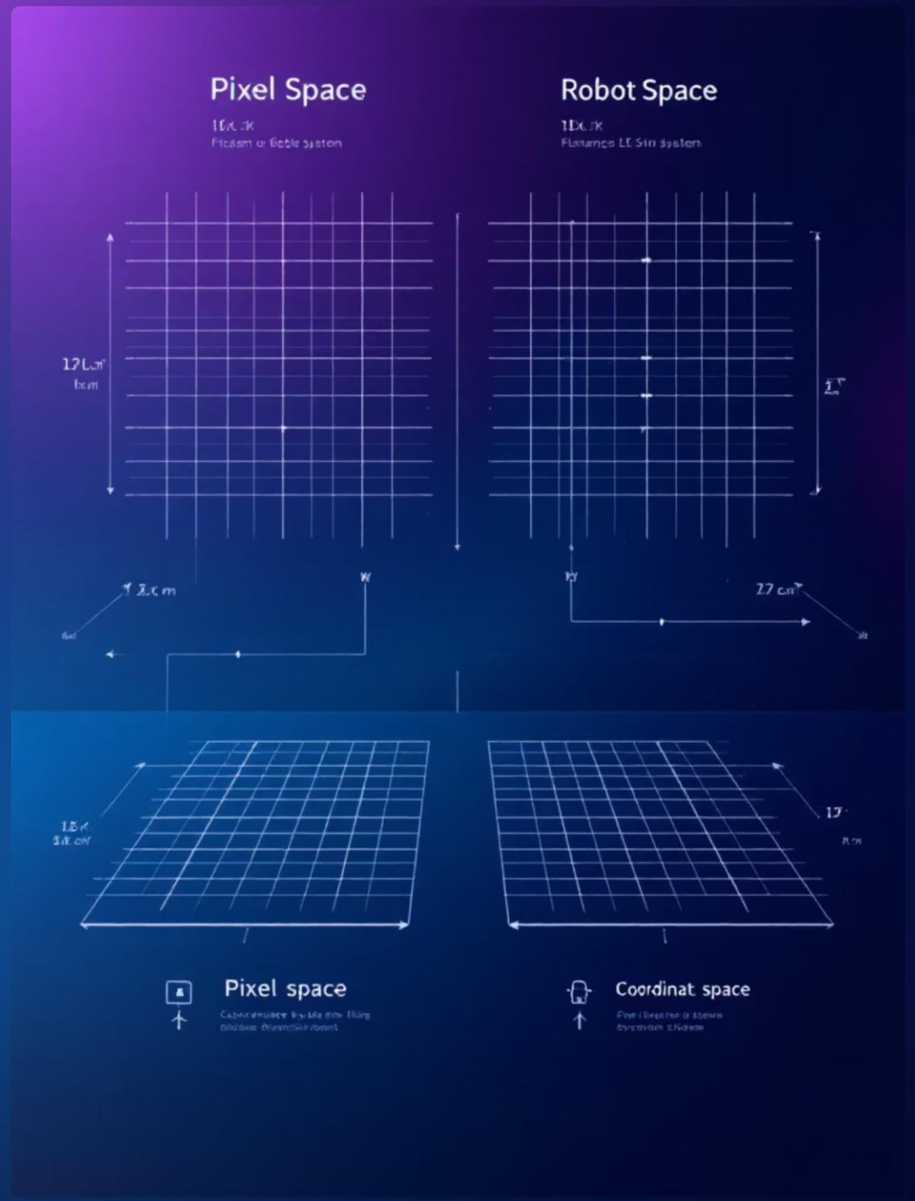
RL Controller (PPO):

- Trained with Stable Baselines 3
- Mean error: 3.5mm (1-5mm band)
- 200 test episodes





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System Integration

Coordinate Transformation:

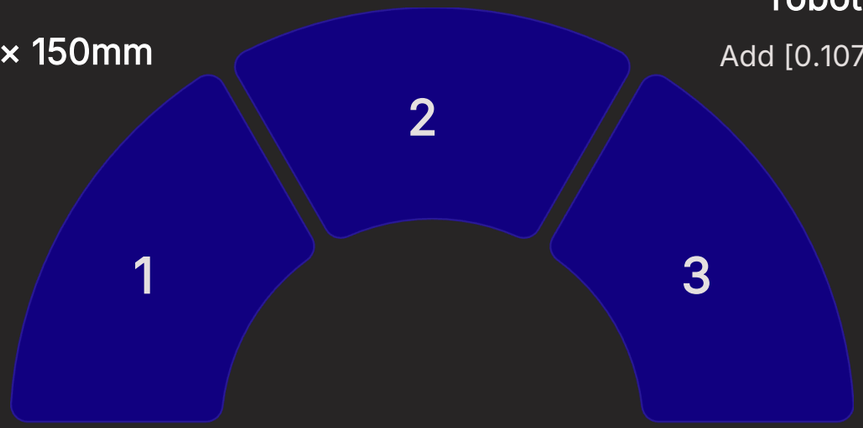
Millimetre Conversion

150mm / plate_pixels = mm per pixel

Add plate position in robot workspace

Add [0.10775, 0.062, 0.1195]

Plate is 150mm × 150mm



Result

Success rate: 100%

Mean error: 0.95mm

Same accuracy as standalone PID testing

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System Performance

Benchmarking: 10 Plates, 50 Plants



Success Rate

50/50 plants inoculated



Mean Error (mm)

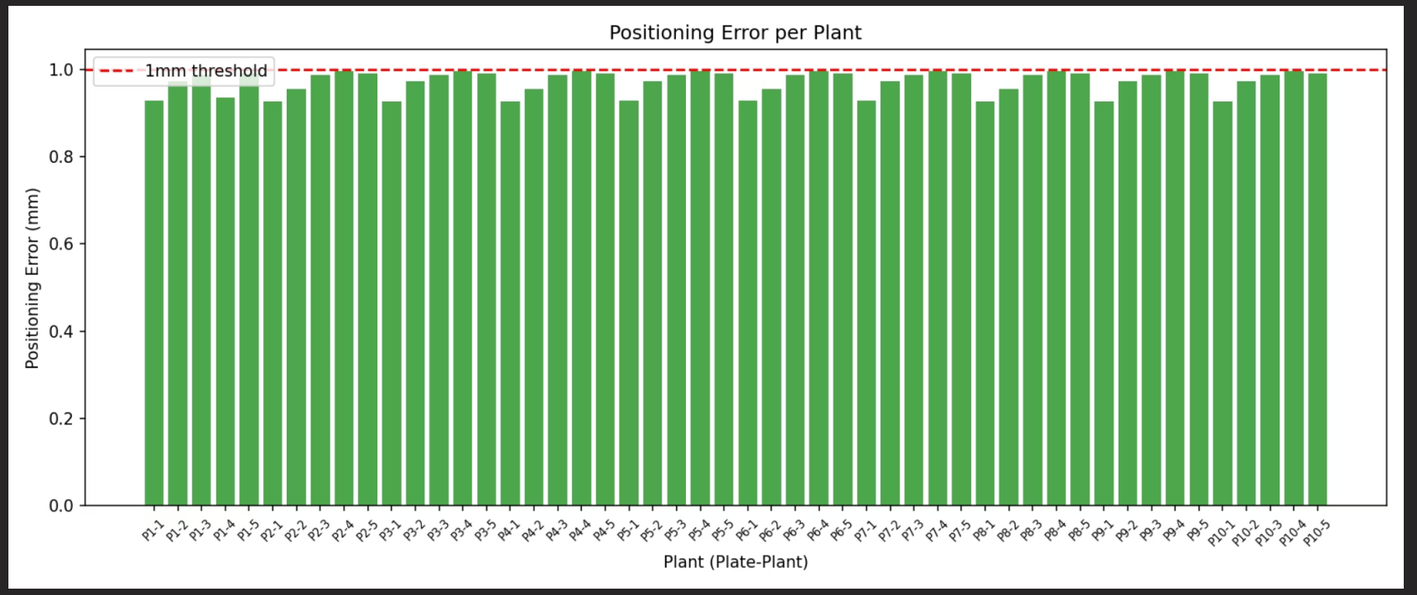
All within 1mm threshold



Seconds per Plate

Time Breakdown

- CV pipeline: 12.5s (39%)
- PID movement: 19.7s (61%)
- Coordinate conversion: <0.1s

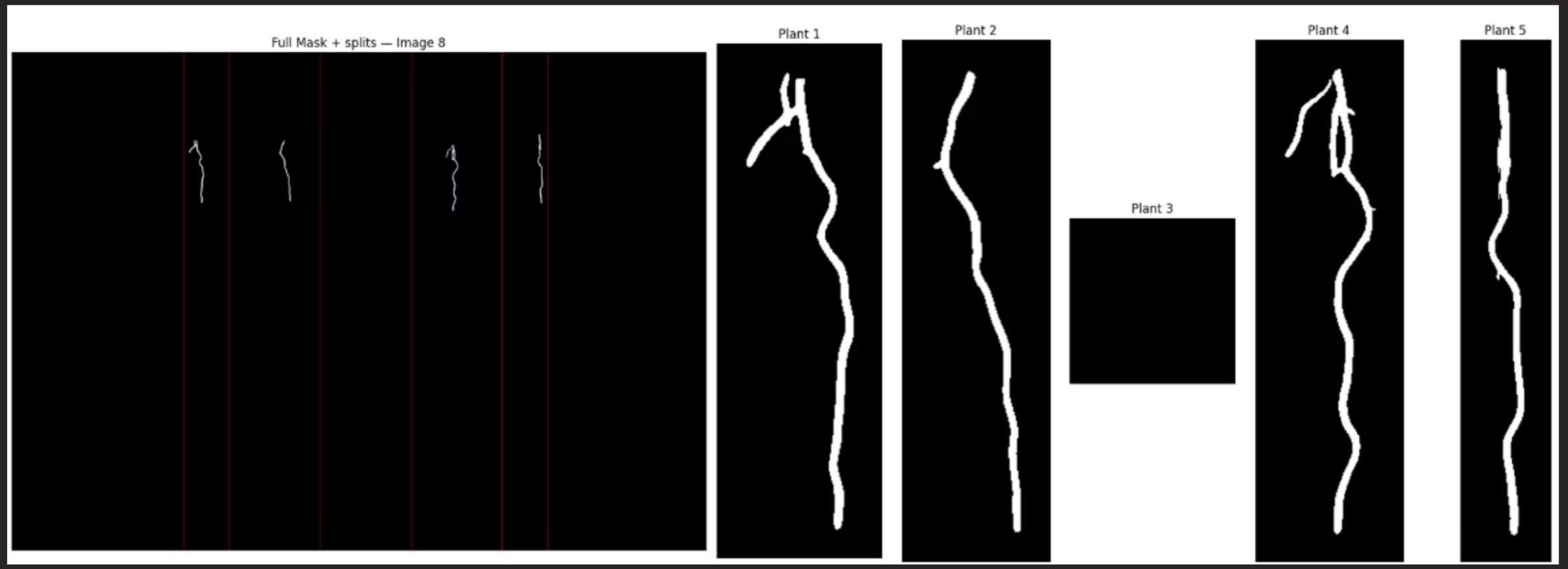


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Iteration 1: Plant Slot Assignment

Problem: When a plate has only 3-4 plants, empty slots were assigned to wrong positions. Plant 3 empty but Plant 5 got assigned as empty instead.

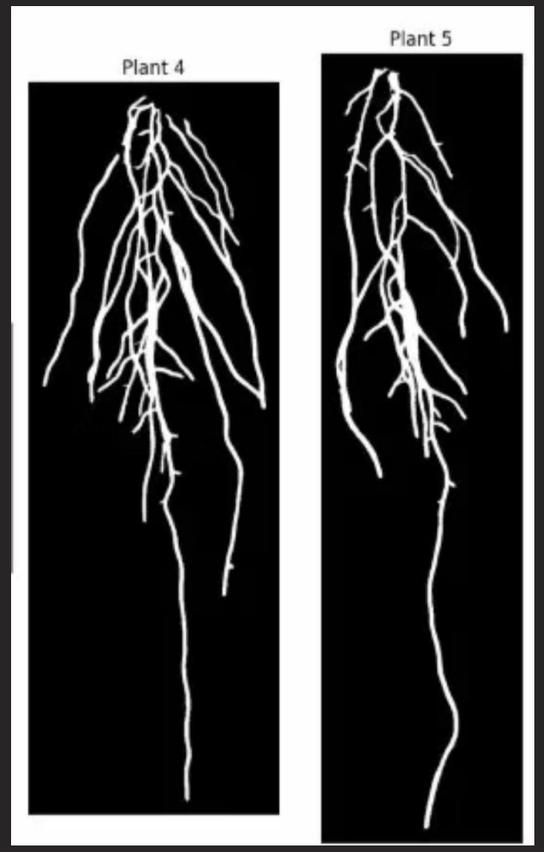
Result: sMAPE dropped from 57% → 24%



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Iteration 2: Primary Root Detection

Result: sMAPE dropped from 24% → 4.1%



Problem: Used highest and lowest skeleton points as primary root. But sometimes the highest or lowest point was a lateral root, not the primary root.

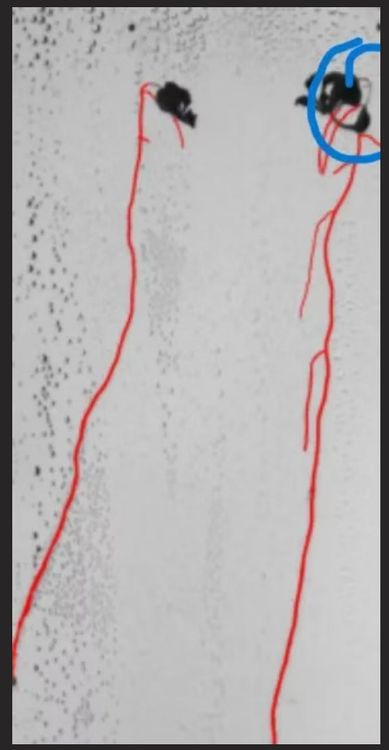
Result: sMAPE dropped from 24% → 4.1%

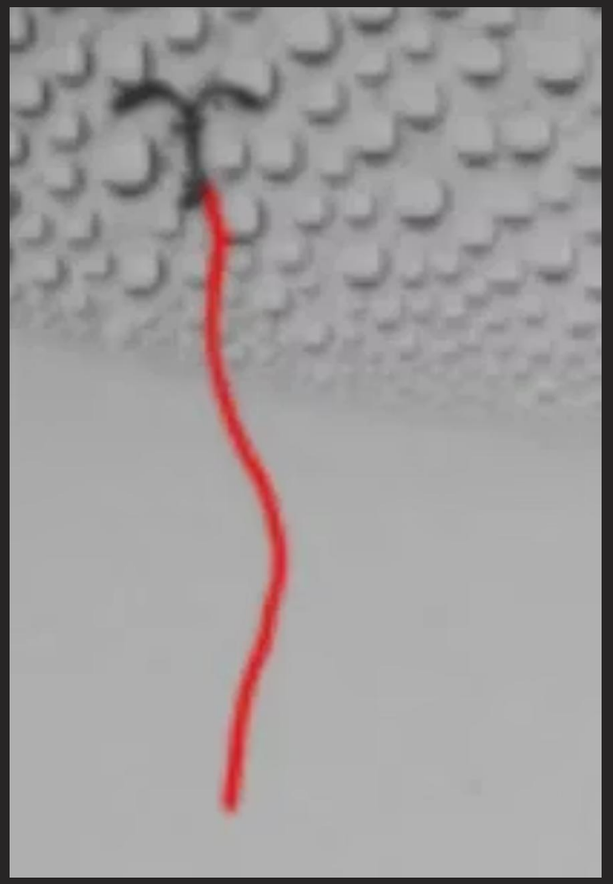
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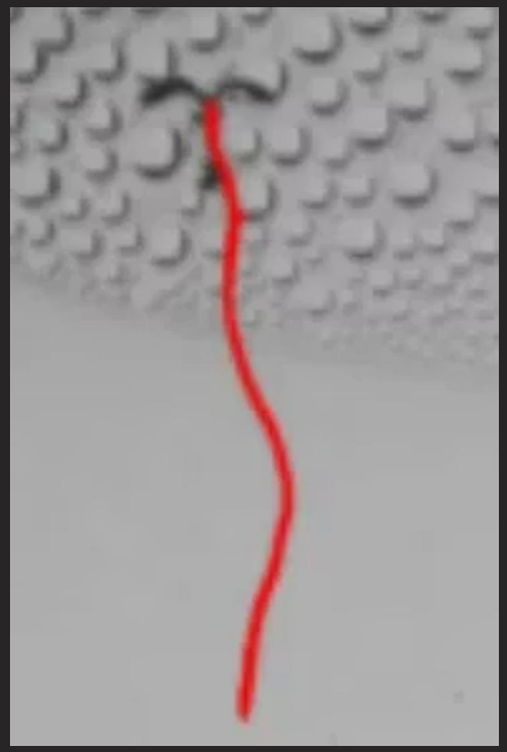
Iteration 3: Root Detection Through Shoots

Problem: Some roots pass through the shoot area. The model could not predict roots there because shoots have no pixels and roots bend near the top.

Result: sMAPE dropped from 4.1% → 3.4%







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Assumptions & Limitations

Assumptions

- Each dish has exactly **5 plants** in a horizontal row
- Plate is **150mm × 150mm** (square)
- Plate position in robot is **known and fixe**

Limitations

- Heavy condensation on plates reduces root detection quality
- Model does not always detect the full root length
- No real-time feedback
- Pipette cannot reach plate surface in simulation (hovers above)
- Plate position is hardcoded — needs calibration for real robot
- CLAHE improves root detection but makes junction look like roots

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Next Steps & Summary

Component	Result
Root Segmentation	F1 = 0.85
Kaggle sMAPE	9.36% (under 10%)
PID Controller	under 1mm error
System Integration	100% success within 1<mm
Speed	32s per plate

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